

Manual testing

What is Software Testing?

- **Software Testing** is a part of software development process.
- **Software Testing** is an activity to detect and identify the defects in the software.
- The **objective of testing** is to release **quality product** to the client.

Software Quality

- **Quality:** Quality is defined as justification of all the requirements of a customer in a product.
 - **Note:** Quality is not defined in the product. It is defined in the customer's mind.
- **Quality software is reasonably**
 - Bug-free.
 - Delivered on time.
 - Within budget.
 - Meets requirements and/or expectations.
 - Maintainable.



Product Vs Project

- If software application is developed for specific customer based on the requirement then it is called **Project**.
- If software application is developed for multiple customers based on market requirements then it called **Product**.

Why do we need testing?

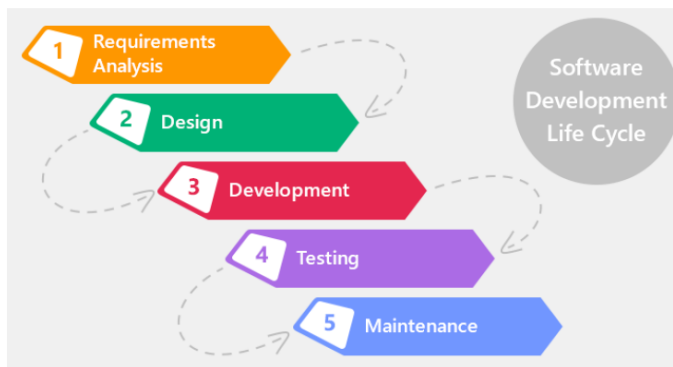
- Ensure that software is **bug free**.
 - Ensure that **system meets** customer requirements and software specifications.
 - Ensure that system meets **end user expectations**.
 - Fixing the bugs identified after release is more expensive.
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- **Error:** Any incorrect human action that produces a problem in the system is called an error.
 - **Defect/Bug:** Deviation from the expected behavior to the actual behavior of the system is called defect.
 - **Failure:** The deviation identified by end-user while using the system is called a failure.

Why the software has bugs normally?

- Miscommunication or no communication
- Software complexity
- Programming errors
- Changing requirements
- Lack of skilled testers

Software Development Life Cycle (SDLC)

- SDLC, Software Development Life Cycle is a process used by software industry to design, develop and test software's.



Waterfall Model(Advantages & Disadvantages)

Advantages of Waterfall Model

- Quality of the product will be good.
- Since Requirement changes are not allowed , chances of finding bugs will be less.
- Initial investment is less since the testers are hired at the later stages.
- Preferred for small projects where requirements are feezed.

Disadvantages of Waterfall Model

- Requirement changes are not allowed.
- If there is defect in Requirement that will be continued in later phases.
- Total investment is more because time taking for rework on defect is time consuming which leads to high investment.
- Testing will start only after coding.

The goal of a software tester is to find bugs, find them as early as possible, and make sure they get fixed.

WHY TESTING IS NECESSARY?

If any level of testing cannot declare that there is no defect in the product, then Why is it required at all?

Development people assume

that whatever they have developed is as per customer requirements and Will always work. But, it is imperative to create real-life scenario and undertake actual execution of a product at each level of software building (including system level) to assess whether it really works or not.

Developers may have excellent skills of coding but integration issues

can be present when different units do not work together, even though they work independently

One must bring individual units together and make the final product, as some defects may be possible when the sources are developed by people sitting at different places.

The primary role Of software testing is not to demonstrate the correctness of

software product, but to expose hidden defects so that they can be fixed. Testing is done to protect the common users from any failure of system during usage.

Testing is a process Of demonstrating that errors are not present in the product,

This approach is used in acceptance testing where if the application meets acceptance criteria, then it must be accepted by the customer,

Testing gives number of detects present which indirectly gives a measurement of software quality.

More number of defects Vindicate bad software and bad processes Of development,

software bug occurs when

1. The software doesn't do something that the product specification says it should do.
2. The software does something that the product specification says it shouldn't do.
3. The software does something that the product specification doesn't mention

The number one cause of software bugs is the specification

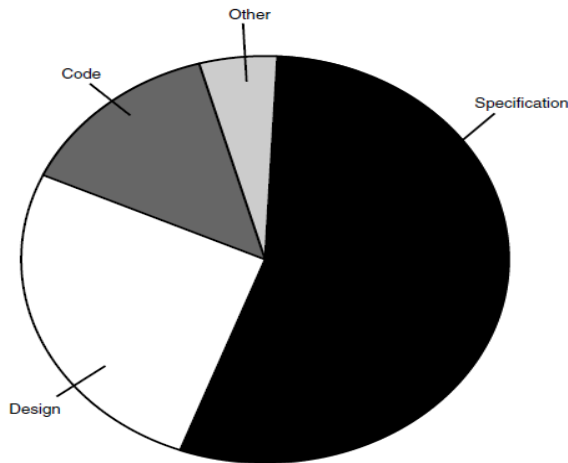


FIGURE 1.1

Bugs are caused for numerous reasons, but the main cause can be traced to the specification.

The bugs are caused for many reasons but main one is Specification.

specifications are the largest bug producer

it's constantly changing, or it's not communicated well to the entire development team.

Planning software is vitally important. If it's not done correctly, bugs will be created.

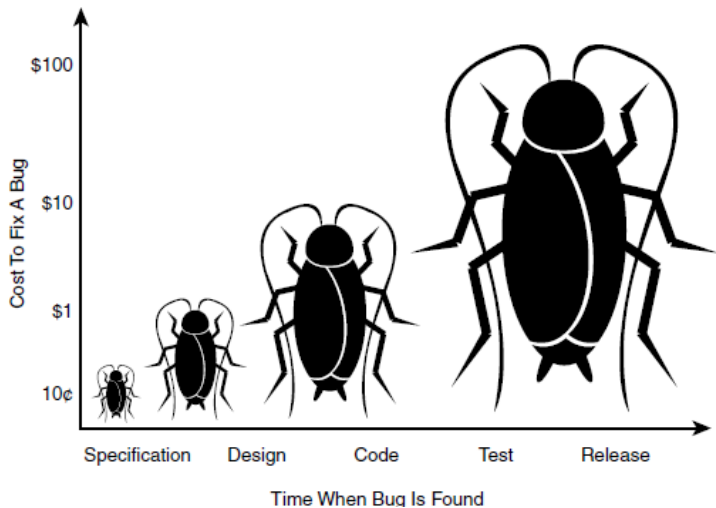
The next largest source of bugs is the design if not designed well then bugs are occurred

NOTE

There's an old saying, "If you can't say it, you can't do it." This applies perfectly to software development and testing.

This means that wrong design ,, more the bug

The Cost of Bugs



The costs are logarithmic—that is, they increase tenfold as time increases. A bug found and fixed during the early stages when the specification is being written might cost next to nothing, or 10 cents in our example. The same bug, if not found until the software is coded and tested, might cost \$1 to \$10.

But If a customer finds it, the cost could easily top \$100.

Example

root cause of the problem was that the software wouldn't work on a very popular PC platform. If, in the early specification stage, someone had researched what PCs were popular and specified that the software needed to be designed and tested to **work on those configurations**, the **cost** of that effort would have been **almost nothing**.

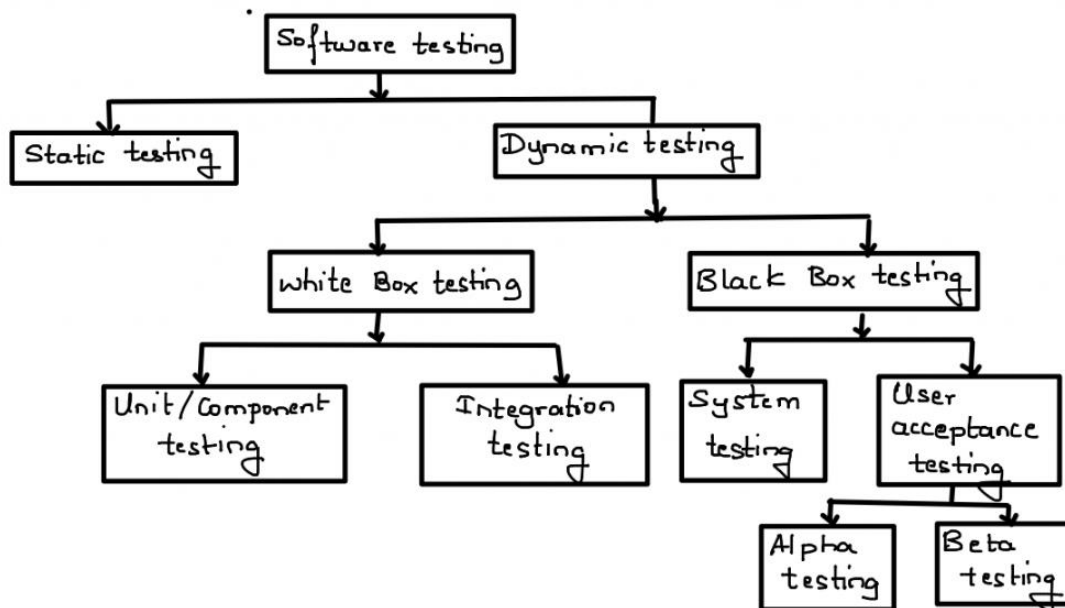
If that didn't occur, a backup would have been for the software testers to collect samples of the popular PCs and verify the software on them. They would have found the bug, but it would have been more expensive to fix because the software would have to be debugged, fixed, and retested.

The goal of a software tester is to find bugs, find them as early as possible, and make sure they get fixed.

Software testing- it is process to test an application to find out error in it.
 Checking the software is ok.
 The goal of software tester to find bug.
 verifying and validating that a software or application is bug free

Testing Types

1. **Manual testing** :- Manual testing includes testing a software manually, i.e., without using any automated tool or any script
2. **Automation Testing**: Automation testing, which is also known as Test Automation, is when the tester writes scripts and uses another software to test the product. This process involves automation of a manual process. Automation Testing is used to re-run the test scenarios that were performed manually, quickly, and repeatedly.



Static Testing Techniques

Analysis of a program carried out without executing the program

Done during verification process i.e before development

As we know 85% errors found in design phase

Is code is tested in static testing ? "No" || the documentation is tested

Software development starts, continues, and ends with documentation

Early documentation → is used to define the software to be built.

Later documentation covers → the software training, installation, and operation (user guides)..

Static = not while running

The primary goal of static testing is reduce defect by reducing defects in the

documentation from which the software is developed

Review : is type of static testing | done before execution

Review is a process or meeting during which a work product or set of work products, is presented to managers, users, customers, or other interested parties for comment or

walkthrough Review :

- It is not a formal process
- It is led by the authors
- Author guide the participants through the document according to his or her thought process to achieve a common understanding and to gather feedback.
- Useful for the people if they are not from the software discipline, who are not used to or cannot easily understand software development process.

Inspection Review:

It is the most formal review type

It is led by the trained moderators

During inspection the documents are prepared and checked thoroughly by the reviewers before the meeting

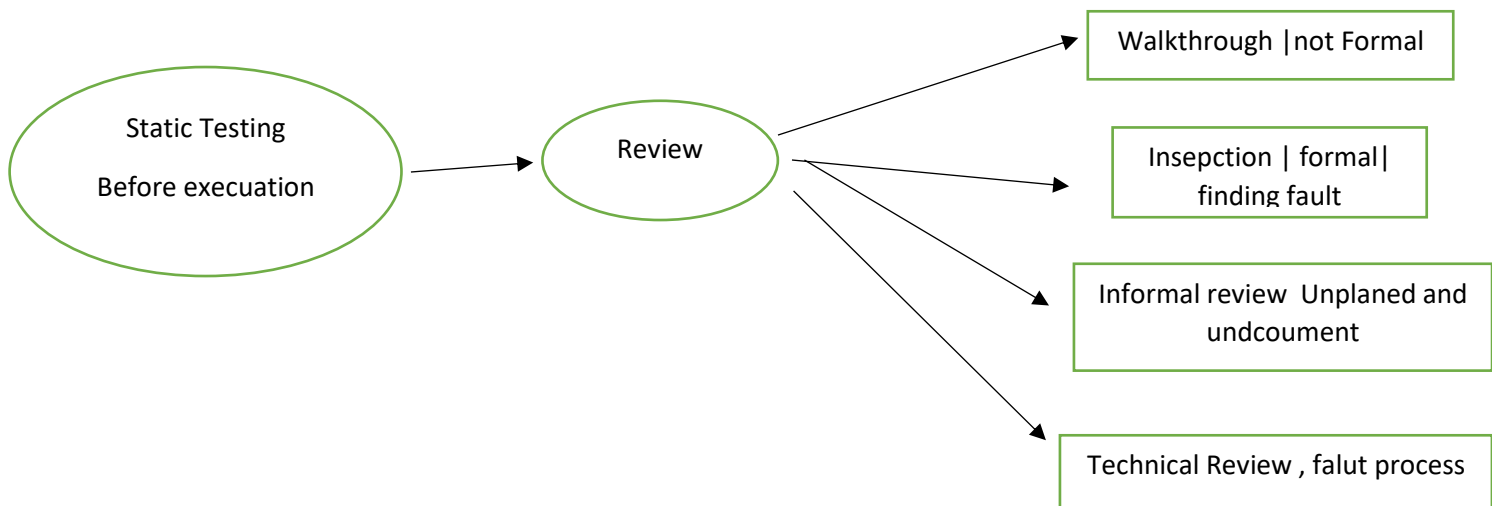
Informal Review

Unplanned and Undocumented

Technical Review

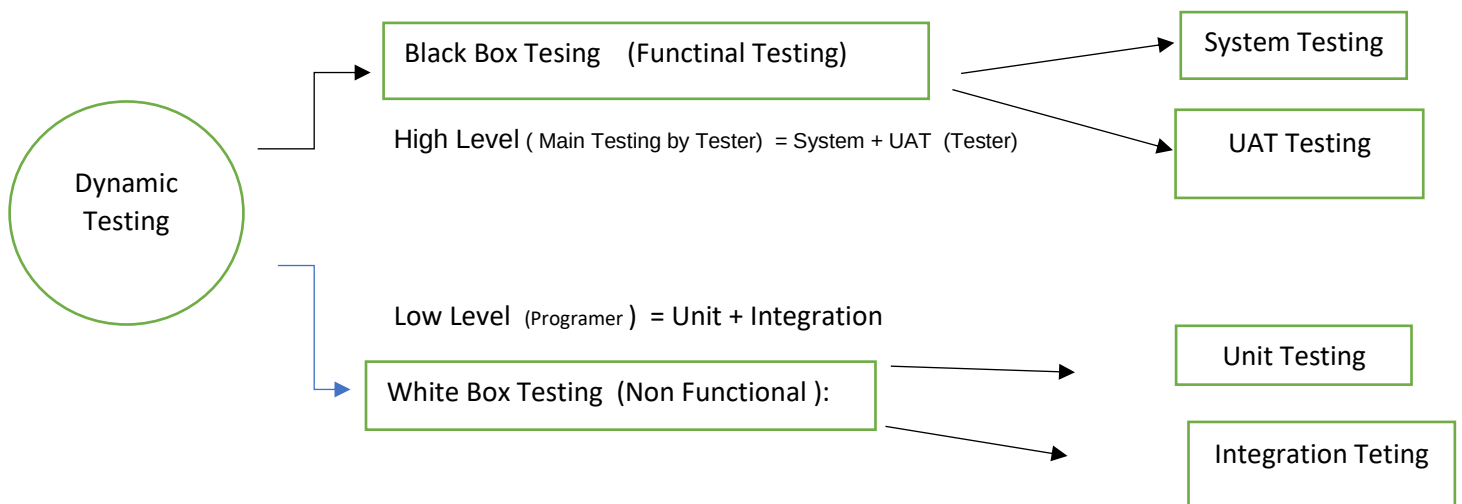
Documented || Defined fault detection process || Includes peers and technical experts

No management participant



Dynamic Testing Techniques

The process of evaluating a system or component based upon its behaviour during execution.



Levels Of Testing

1. Unit Testing :- in unit testing individual component of software tested. The purpose of this testing is that each module is working properly
It focuses on the smallest unit of software design
(done by developer by using sample input and observing its sample output)

Eg. In a program we are checking if loop, method or function is working fine

2. Integration Testing: in integration Testing individual units are combined and tested as group (developer)

- (i) Top-down
- (ii) Bottom-up
- (iii) Sandwich
- (iv) Big-Bang

Main Purpose of Integration Testing :

To check modules are communicating each other as DFD Data Flow Diagram which is specified in TDD (Technical Document Diagram)

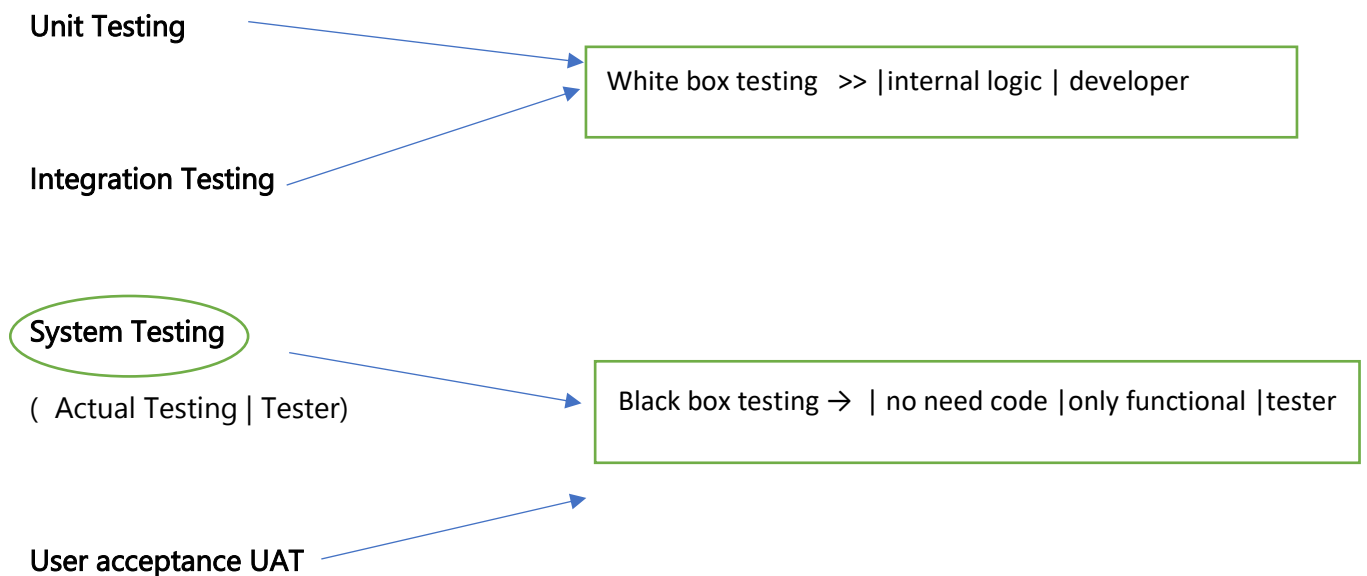
3. System Testing : in this testing we can test whole application (complete / integrated software is tested) done by tester

4. Acceptance Testing : a level of software Testing in which software is tested for user acceptance

UAT done at client location where software is actually used

→ Alpha Testing : done by tester in company in presence of customer

→ Beta Testing : done by customer to check software is ok , satisfy requirement



Testing Types

Functional testing

is the process through which QAs determine if a piece of software is acting in accordance with pre-determined requirements. It uses black-box testing techniques, in which the tester has no knowledge of the internal system logic. **Functional testing is only concerned with validating if a system works as intended**

Functional Testing : Testing what system does

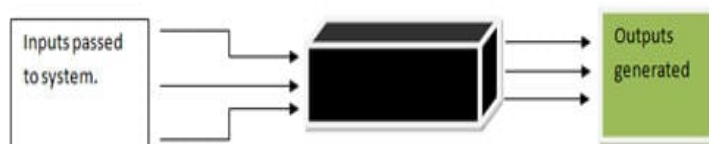
Functional testing is a type of black-box testing. *“does this actually work?”*

The ultimate goal of functional testing is to ensure that software works according to specifications and user expectations

input values

test cases

Compare actual and expected output



Eg. Login functionality, registration functionality

1. Non Functional Testing :

- Load testing
- Reliability
- The readiness of a system
- Usability testing

Eg. A practical example would be: checking how many people can simultaneously check out of a shopping basket

2. Black Box Testing : (without code) High level

Black box testing is that kind of software testing you can do when you do not have the source code, just the executable code in hand

The testing is done without the internal knowledge of the products

3. White box testing : (with code) low level

Monitoring internal structure . check internal logic . done by developer

4. Smoke testing : - (Testing on newly released build → compulsary requirement)

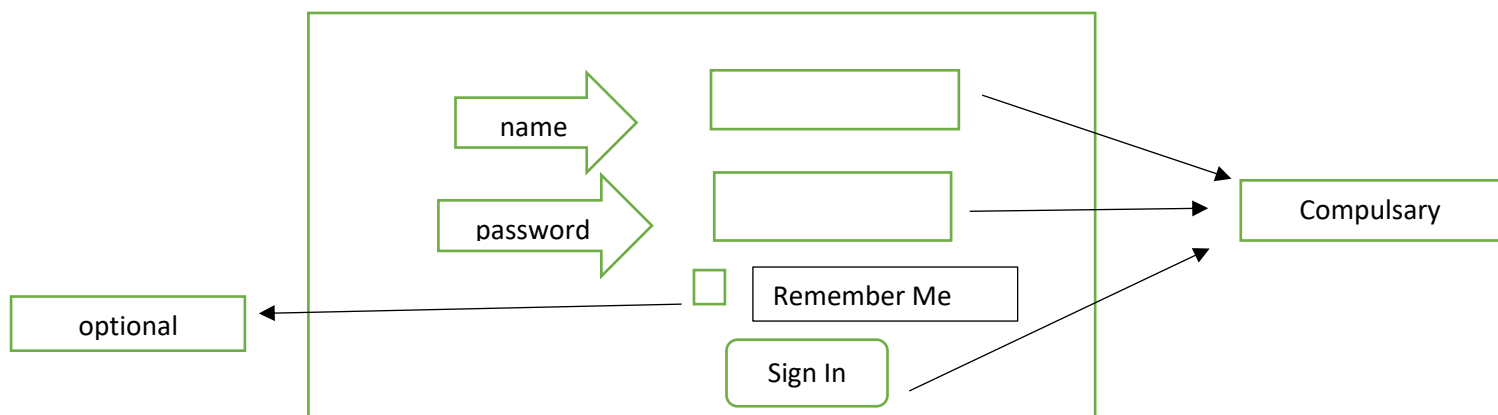
It is first testing on newly released build ... (Build Verification Testing)

Check → the deployed software build is stable or not.

Eg

QC → Build s/w → QA → Testing

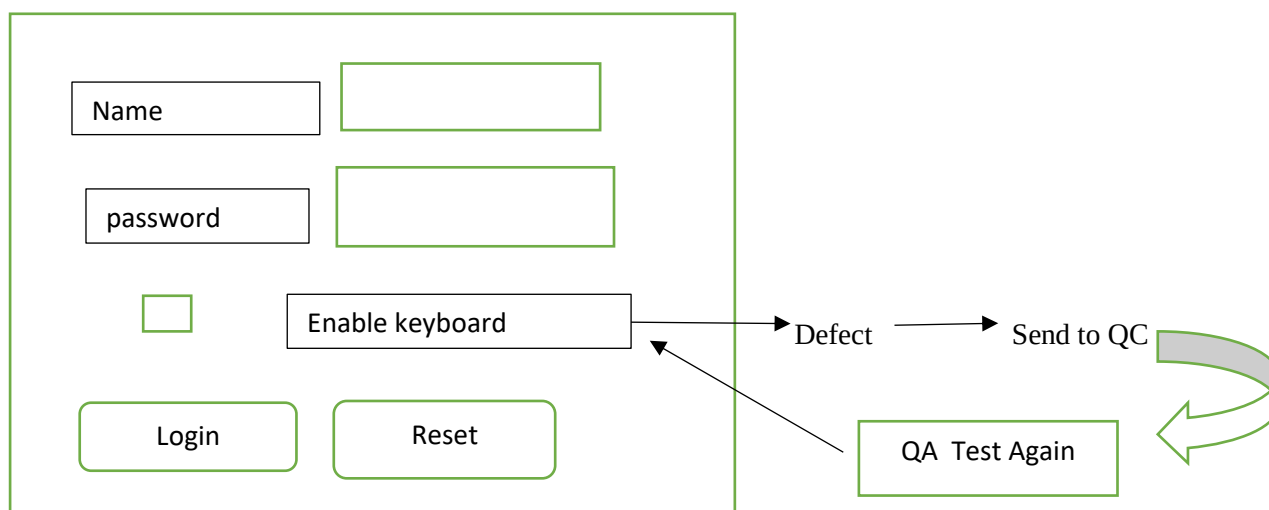
Compulsory Testing → smoke testing



Sanity Testing : (Testing on newly released build → check **Compulsory + optional**)

In above fig all filed are compulsory then it is sanity Testing

Retesting : Testing functionality once again



Regression Testing

(Re- running if code changes) :

It is overall testing when ever new change is occurred .

Re-running functional + non Functional test

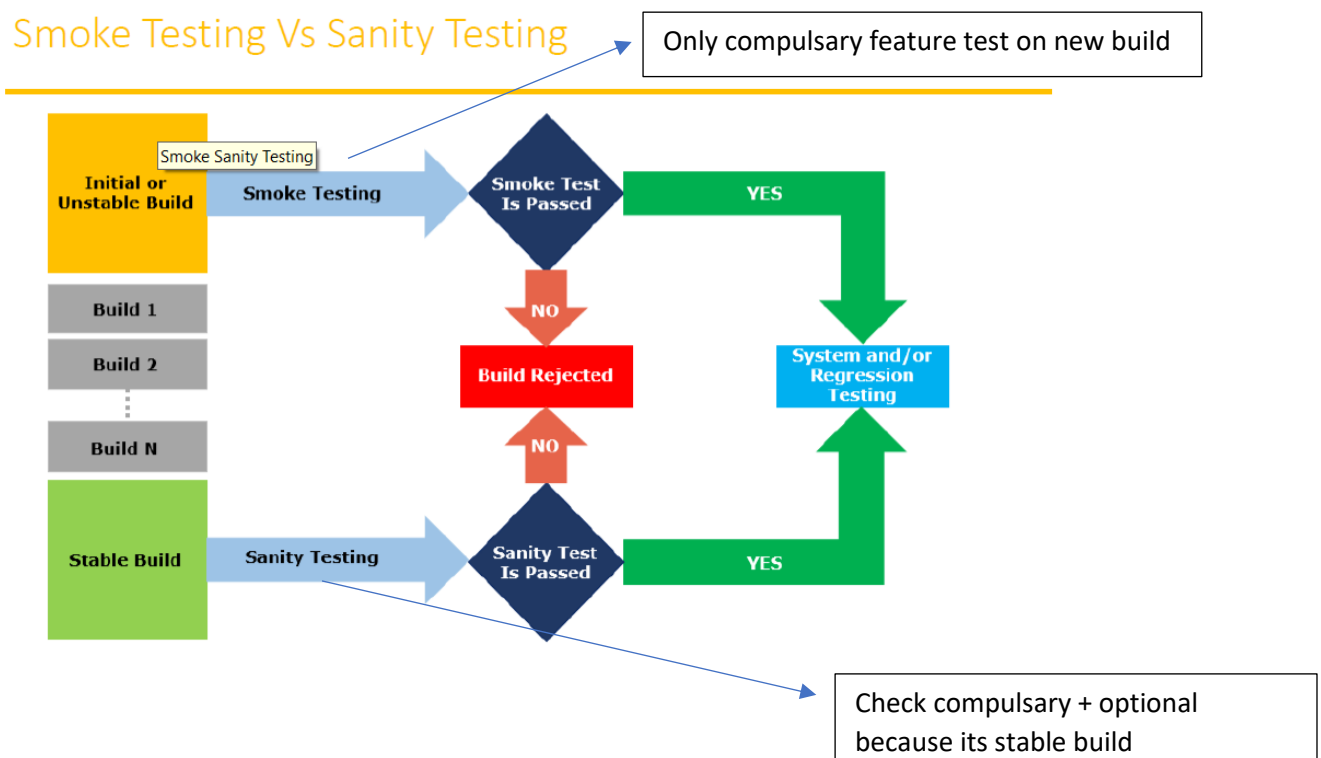
Code change , does not impact existing function.

After change code , is software works Ok.

Every time a new module is added leads to changes in the program.

This type of testing makes sure that the whole component works properly even after adding components to the complete program.

Smoke Testing Vs Sanity Testing



Static Testing :

Review
Inspection
Walkthrough

Test software without executing software (just test documents)

This test done for avoid bugs in early stage (review testing) look and feel

Why Static testing

- Early defect detection and correction
- To get fewer defect at a later stage of testing

Static technique has 3 types

Review : review before development i.e simple document

Read the document, correct or not
 Document s/b correct and complete
 Requirement review , design review , Test plan review
 Review testing can do anybody , manager|developer|tester| coworker etc.

Walkthrough :

its is informal , anytime , not planned , done when ever required
 author of document will explain to their team

Inspection :

Most formal , 3- 8 people in meeting
 Proper meeting , schedule which is intimated by mail

Dynamic Testing : A testing which is done after code development.

The main purpose of dynamic testing is to test software behaviour with dynamic variables

dynamic testing requires code to be executed

static testing → just analyse the code, no need execution

Alpha Tesing : Its is final testing in development

Advantage : immediate solution is possible

Beta Testing : it is 1st testing in client side . it is also called user user acceptance testing UAT

Disadvantage : no immediate solution if defect is found

Installation Testing : providing required resources at client location

It is type of testing in which test engineer check deployment process is successful as per user guideline

Deployment document /user manual : it is document prepared by project manager

Usability Testing : checking application for user friendliness

Monkey Testing : used for game testing, used for random input

To check the application or system will crash

Portability Testing : Developed application Shoud support multiple enviornment

Forced error Testing : to check valid error message will display

QA | QC | QE

QA = Highest Possible quality

quality assurance | process related | high level management

Process designed by QA | Responsible for highest possible quality

QC = quality control | product related | actual tester

QA is process orientated | QC product orientated (that work on product actual tester)

QA = responsible for preventing defect (involves all phase i.e. design coding ..)

QC = responsible for finding defect (involves only testing)

QE = Quality engineer

Responsible for write code for testing (automation engineer)

System Testing

(Actual Testing)

System Testing : GUI Testing | Usability Testing | Functional Testing | non functional

1. GUI Testing :

- : - Testing GUI application , user interface testing
- : - such as menus , check boxes , icon, images
- : - not functional , just look and feel
- : - check size and position of element
- : - image quality, spelling check , alignment
- : - Fonts are understanding or not

2. Usability Testing :

- : - check the easiness of application
- : - helping messages are display if user confuse
- : - check user friendly application or not ?

3. Functional Testing :

- :- check behaviour of application
- :- check database testing (work with database ok ?)
- :- error handling . display error message ok ?
- :- calculation and manipulation

Eg. $5+5 = 15$ (user requirement) => we follow this

$5+5 = 10$ (math calculation)

- :- check text box disable or enable as user requirement
- :- Check database operation DML table, column , records etc



Checking database operation

Black box testing + white box testing = Gray box Testing

4. Non Functional testing :

- :- once functional testing done i.e. s/w work user requirement then do non functional testing
- :- performance testing
 - Load testing – gradually increase the load
 - Stress testing – suddenly increase the load (Eg. Online filling form)
 - Volume testing – how much data handle
- :- security of software
- :- recovery of application
- :- Compatibility testing – work with all platform

End to End Testing:

:- testing overall application after including all module

Eg : login → add customer

→ delete and edit customer

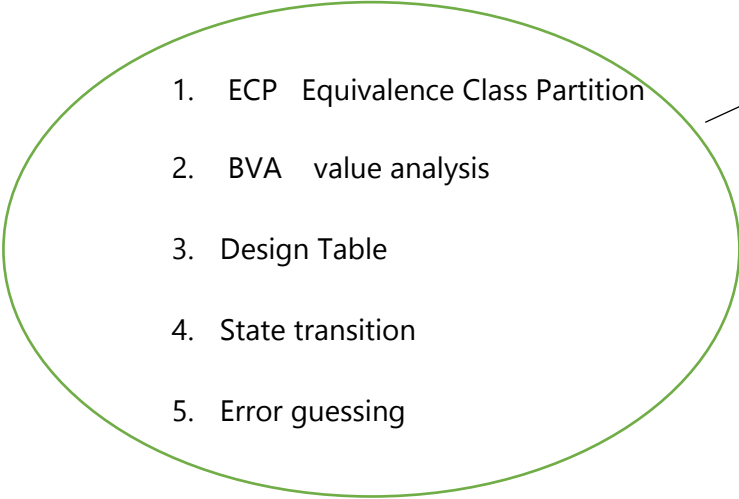
→ logout

Testing all function i.e. add delete edit and logout

Test Case Design Technique:

It helps better design and reduce the number of test case to be executed

Reduce data and more coverage

- 
1. ECP Equivalence Class Partition
 2. BVA value analysis
 3. Design Table
 4. State transition
 5. Error guessing

Black Box Testing
(Just design test case & run)

All this are Test Case Design technique

(all this are in Black Box , Its Main testing)